CHAPTER

5

CONCEPT OF PRODUCTION

Course of Action
Maximum Time = 3:15 Hrs
Maximum Questions = 63

Learning Objectives

The students will learn:

- *Meaning of Production, Product and various types of products.*
- *Production Functions, both short run and long run.*
- *Law of variable proportions and the stages of the law.*
- *The reasons for various stages of the law of variable proportions to occur.*

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Topic-1

Concepts of Production

Concepts covered: Meaning and concept of production and production function, Time period- Short run time period and long run time period. Types of production function–(a). Short run production function, (b) Long run production function. Types of factors–(a) fixed factors, (b) Variable factors. Types of product– (a) Average product, (b) Total Product, (c) Marginal Product.

Revision Notes

- **Production:** Production may be defined as a process through which a firm transforms inputs into output. It is the process of creating goods and services with the help of factors of production or inputs for satisfaction of human wants. In other words, 'transformation of inputs into output' whereby value is added, is broadly called production.
- **Production Function:** Production Function is the functional relationship between physical input and physical output. In production function, only physical inputs are taken into consideration in relation to the physical output and the market prices are not included in it.

Scan to know more about this topic

Production function

$$= f\{L,K\}$$

- **Short-run:** Short-run refers to a period in which output can be changed by changing only variable factors. Fixed factors like plant, machinery, building, etc. cannot be changed.
- **Long-run:** Long-run refers to a period in which output can be changed by changing all factors of production. A firm can change its production techniques, purchase new machinery, etc.
- **Types of Production Function:** There are two types of production function:

(i) **Short Run Production Function or Returns to a Factor:** It is related to short-run, when only one factor of production is variable and others are fixed. Since units of variable factors are increased continuously, so factor ratio changes in this production function.

(ii) **Long Run Production Function or Returns to Scale:** It is related to long-run and all the factors are changed in same ratio. In this case, proportional relation between production and factors of production in the long-run is called Returns to Scale. Here, scale of production changes.

- **Total Product or Total Physical Product:** Total production refers to total output produced by a firm during a given period of time with given number of inputs.
- **Average Product:** Average product refers to output per unit of a variable input. To get Average Product, we divide total product by amount of variable factor.

$$AP = \frac{TP}{L}$$

Concept of Production

Some Basic Concepts

- **Total Physical Product**
  - Total quantity of commodity produced by a firm with given variable output keeping fixed factors constant
  - TPP = AP × L (L is units of variable factor)
- **Average Physical Product**
  - Output per unit of variable input
  - $AP = \frac{TP}{L}$
- **Marginal Physical Product**
  - Addition to TP by additional unit of variable input
  - $MP = \frac{\Delta TP}{\Delta L}$ or $MP = TPn - TP_{n-1}$

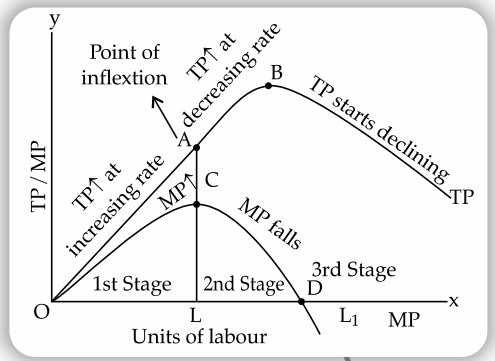
Types

- **Long run production function** (FF↑↓→ VF↑↓) (Return to scale, types)
  - **Increasing Returns to Scale**: 1M + 1L = 100 units; 2M + 2L = 220 units
  - **Constant Returns to Scale**: 1M + 1L = 100 units; 2M + 2L = 200 units
  - **Decreasing Returns to Scale**: 1M + 1L = 100 units; 2M + 2L = 180 units
- **Short-run Production Function** ($\overline{FF}$ → VF↑↓) (Return to a Factor /Law of variable Proportion)

Assumptions

- State of technology is constant
- One factor is variable and others are fixed
- Homogeneous unit of variable factors
- Factors of production are not perfect substitutes
- It is possible to make changes in the factor proportion
- There is a short period of operation

Graphical presentation



Stages

- **First Stage** (Stage of Increasing Returns: MP tends to increase and TP increases at increasing rate)
  - Due to :
    - Optimum combination of factors
    - Fuller utilisation of fixed factors
    - Division of labour & specialisation
    - Volume discount
- **Second Stage** (Stage of Diminishing Returns: MP decreases but remains positive and TP increases at decreasing rate)
  - Due to :
    - Scarcity of resources (Supply of fuel, power, raw material, etc.)
    - Use beyond optimum capacity.
    - Lack of perfect substitution beween factors up to certain limit.
- **Third Stage** (Stage of Negative Returns : MP becomes Negative, TP Starts Declining)
  - Due to :
    - Over crowding
    - Management Problem

Schedule

FF land in acre	VF labour (Unit)	TP (Unit)	MP (Unit)	Returns to Variable factor	Stage of Productions
1 1 1	1 2 3	5 15 30	5 10 15	TP↑ at increasing rate MP↑	Stage -I (Increasing Returns)
1 1 1	4 5 6	40 45 45	10 5 0	TP↑ at decreasing rate MP↓ (but remains positive)	Stage -II (Decreasing Returns)
1 1	7 8	40 30	–5 –10	TP↑ MP↓ (get Negative)	Stage -III (Negative Returns)

Trace the Mind Map

- First Level
- Second Level
- Third Level

* L = Labour
* C= Capital
* FF = Fixed factors
* VF = Variable Factors

- **Marginal Product:** Marginal product is the change in Total Product resulting from the use of one more (or one less) unit of the variable input, keeping all other inputs constant.

$$MP = \frac{\Delta TP}{\Delta N} \quad \text{OR} \quad MP_n = TP_n - TP_{n-1}$$

Example: Complete the following schedule:

Units Produced	TP (in ₹)	AP (in ₹)	MP (in ₹)
1	50		
2			70
3		70	
4	240		

Ans.

Units Produced	TP (in ₹)	AP (in ₹)	MP (in ₹)
1	50	50	–
2	120	60	70
3	210	70	90
4	240	60	30

Key Terms

- **Variable factors:** Its production are those factors of production, the application of which changes with the change in output.
- **Fixed factor:** Fixed factors of production are those factors of production, the application of which does not change with the change in output.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. Define total product? [OEB]

Ans. Total product can be defined as the total output produced by applying various units of variable factor and fixed factor.

Q.2. What is the meaning of production? [OEB]

Q.3. What do you mean by Average Product? [OEB]

Ans. Average product refers to output per unit of a variable input. To get Average Product, we divide total product by amount of variable factor.

$$\text{AP} = \frac{TP}{L}$$

Q.4. What will be the average product if the Total Product is 250 units, with 10 units of labour? [OEB]

Ans. $\text{AP} = \frac{TP}{L} = \frac{250}{10} = 25$

Q.5. If the TP_5 is 10 and TP_4 is 12, what is the MP? [OEB]

Ans. $MP = TP_5 - TP_4 = 10 - 12 = -2.$

Q.6. State giving reason whether the following statement is true or false: [OEB]

(a) "All the factors become variable in the long run."

Ans. True

Reason: The given statement is true as there exists both fixed factors and variable factors in the short run but in the long run only variable factors exists because all factors can be changed in the long run.

(b) "In short run production function, both the fixed and variable factors become variable."

Ans. False

Reason: The given statement is false because in short run production function, both the fixed factors and variable factors exists. As fixed factors cannot be changed in the short run.

(c) "Average product can be referred to as per unit of variable factor of production."

Ans. True

Reason: The given statement is true because average product can be referred to as the per unit of variable factor of production.

Average product = Total Product/ Units of variable factor

(d) "Marginal product can be referred to as a change in average product when an additional unit of variable factor is employed."

Ans. False

Reason: The given statement is false because Marginal Product can be referred to as the change in total product when an additional unit of variable factor is employed.

2 Marks Questions

Q.1. What is short run production function? Explain how is short run production function different from long run production function. [ISC 2017]

Ans. Production function expresses the relation between physical input and physical output of a firm given the state of technology. It is a purely **technical relationship** that connects factor- inputs and outputs.

$Q_x = f(F_1, F_2, F_3, \ldots F_x)$

Q_x = quantity of commodity x

$F_1, F_2, F_3 \ldots F_x$ = different factor inputs

This question is for practice and its solution is available at the end of the chapter

Differences:

Short-run production function: It refers to a situation when only one input is variable and all other inputs are assumed to be constant. The behaviour of output when only one input is changed and other inputs are fixed are termed as **returns to a factor**. It is studied as law of variable proportions or laws of returns.

Long –run production function: It refers to a situation when all the inputs are variable. The way in which output varies when all inputs in production of a commodity are changed simultaneously in the same proportion are studied. It is studied as **returns to scale**.

[ISC Marking Scheme, 2017]

Examiner's Comment

- Most of the candidates answered this part correctly. A few candidates did not write the difference with respect to fixed and variable factors.

Answering Tip

- Learn the role of inputs in different time periods. Students should also understand that short run refers to the law of variable proportions.

Q.2. What is production function? [ISC 2020]

OR

What is meant by production function? [ISC 2015]

Q.3. What is meant by actual earning of a factor? [ISC 2010]

Ans. Actual earning of a factor is the amount of money earned or received to the various factors of production in exchange of rendering their factor services. The various factors of production are land, labour, capital and entrepreneur. The earnings of these factors are rent, wages, interest and profit respectively.

Q.4. Mahima said to Siya,"The time period in which all the factors of production are variable is called as short run." Analyse the above statement as said by Mahima and explain whether this is correct or not. [OEB]

Ans. Mahima's statement is not correct as the time period in which all the factors of production becomes variable is called as long run. And the time period in which one of the factor is fixed and other factors are variable is called as short-run.

Q.5. Mohan said to Seema that the following relationship exist between TP, AP and MP: [OEB]

AP = TP/Units of fixed factor

MP = TP(n-1) - TPn

Analyse the above statement as said by Mahima and explain whether it is correct?

Ans. The given statement is false and correct one is written as follows:

AP =TP/ Units of variable factor

MP = TPn - TP(n-1)

3 Marks Questions

Q.1. Give any two differences between returns to a factor and returns to scale. [ISC 2023]

OR

Discuss two differences between returns to scale and returns to a variable factor. [ISC 2019]

Ans. Differences between returns to scale and returns to a variable factor.

	Returns to Scale		Returns to Variable Factor
1.	Quantity of **all factor inputs is changed.**	1.	Quantity of **only one factor is changed.**
2.	Operates in the **long run.**	2.	Operates in the **short run.**
3.	Factor proportion **remains the same.**	3.	Factor proportion **changes**.
4.	**Change in scale** of production.	4.	**No change in the scale** of production.

[ISC Marking Scheme, 2019]

Examiner's Comment

- Majority of the candidates were unable to write the differences between returns to scale and returns to a variable factor. Several candidates did not write about the production function of these two time periods. Most of the candidates wrote only long run and short run as a difference and did not mention the other relevant differences.

Answering Tip

- Learn all relevant differences between returns to scale and returns to a variable factor, laying stress about the time period and their production function.

This question is for practice and its solution is available at the end of the chapter

Q.2. Fill the blanks in the table given below: [ISC 2017]

No. of workers	TP	AP	MP
1.	--	150	--
2.	230	--	--
3.	--	--	120

Ans.

No. of workers	TP = AP × No. of workers	$AP = \frac{TP}{\text{No. of Workers}}$	$MP = TP_n - TP_{n-1}$
1.	150	150	---
2.	230	115	80
3.	350	116.66	120

[ISC Marking Scheme, 2017]

Examiner's Comment

- By and large this question was answered well by candidates. Only a few candidates committed mistakes by writing zero for M.P. in the first row.

Answering Tip

- Practice a number of similar problems.

Q.3. Differentiate between fixed factor of production and variable factor of production. [ISC 2016]

Ans. The difference between fixed factor of production and variable factor of production are:

S. No.	Fixed Factor of Production	Variable Factors of Production
1.	It remains fixed in the short run with the change in output.	It changes from time to time as per the production with the change in output.
2.	Land in the short run.	Labour in the short run
3.	No factor is fixed in the long run.	All factors are variable in the long-run.

Examiner's Comment

- Fixed factor – remains constant with "output" variable factor- changes with "output" – here Output was the key word. A number of candidates did not mention "output". In several cases, fixed factor and variable factor were confused with fixed cost and variable cost.

Answering Tip

- Differentiate the two factors with reference to change in "output".

Q.4. Differentiate between short-run production function and long-run production function.

[OEB]

4 Marks Questions

Q.1. Complete the following table: [OEB]

Units Produced	Total Product (in ₹)	Average Product (in ₹)	Marginal product (in ₹)
1	200	__	__
2	___	__	180
3	___	180	__
4	630	__	__

Ans.

Units Produced	Total Product (in ₹)	Average Product (in ₹)	Marginal product (in ₹)
1	200	200	__
2	380	190	180
3	540	180	160
4	630	157.5	90

Q.2. Complete the following table: [OEB]

Units Produced	Total Product (in ₹)	Average Product (in ₹)	Marginal product (in ₹)
1	20	__	__
2	__	15	10
3	45	__	__
4	__	__	20
5	78	__	__
6	88	__	10

Ans.

Units Produced	Total Product (in ₹)	Average Product (in ₹)	Marginal product (in ₹)
1	20	20	__
2	30	15	10
3	45	15	15
4	65	16.25	20

This question is for practice and its solution is available at the end of the chapter

5	78	15.6	13
6	88	14.67	10

Q.3. Complete the following table: [OEB]

Units Produced	Total Product (in ₹)	Average Product (in ₹)	Marginal product (in ₹)
1	150	150	__
2	300	__	__
3	__	__	180
4	__	__	120
5	700	__	__

Ans.

Units Produced	Total Product (in ₹)	Average Product (in ₹)	Marginal product (in ₹)
1	150	150	__
2	300	150	150
3	480	160	180
4	600	150	120
5	700	140	100

Q.4. Complete the following schedule: [OEB]

Units Produced	TP (in ₹)	AP (in ₹)	MP (in ₹)
1	100	__	__
2	__	__	140
3	__	140	__
4	480	__	__

Ans.

Units Produced	TP (in ₹)	AP (in ₹)	MP (in ₹)
1	100	100	–
2	240	120	140
3	420	140	180
4	480	120	60

6 Marks Questions

Q.1 What are the assumptions of the production function? [OEB]

Ans. Some of the assumptions that the analysis of the production function is based on are:

(i) Both the input and the output are perfectly divisible.

(ii) Limited ability to substitute one factor for another because labour and capital are insufficient replacements.

(iii) The state of technology is unchanging.

(iv) The business maximizes the efficiency of its inputs.

(v) It has to do with a specific interval of time.

(vi) If one of the variable factors changes, the level of output will also change.

Q.2. Let the production function of a firm be $Q=2L^2K^2$. Find out the maximum possible output that the firm can produce with 5 units of L and 2 units of K. What is the maximum possible outputs that the firms can produce with zero unit of L and 10 units of K? [OEB]

Ans. (a) $Q=2L^2K^2$

L= 5 units of labour

K= 2 units of capital

Putting these values in equation (1)

$Q=2(5)^2(2)^2$

$=2(25)(4)$

Q=200 Units

(b) If L=0 units and K=100 units

Putting these value in equation (1)

$Q=2(0)^2(100)^2$

Q=0 Units

Topic-2 Law of Variable Proportions

Concepts covered: Concept of Law of variable proportion, Stages and assumptions of law of variable proportion, Relationship between TP, AP and MP, Causes of the operation of law of variable proportion, Law of diminishing returns to factor.

Revision Notes

- **Law of Production or Law of Variable Proportion:** Law of Variable Proportion states that as more and more of the variable factor is combined with the fixed factor, a stage must ultimately come when marginal product of the variable factor starts declining.
- **Assumptions of the Law of Variable Proportion**: **(i)** Short run, **(ii)** Factors can be substituted mutually, **(iii)** Technology is constant, **(iv)** All units of variable factors are homogeneous.
- **Stages of Law of Variable Proportion:** According to this law, there are three stages of production:

(i) First Stage: TP increases at an increasing rate and MP rises till it reaches its maximum point.

(ii) Second Stage: Total product increases at a decreasing rate and reaches maximum, and MP becomes zero.

Scan to know more about this topic

Law of Variable Proportions

(iii) Third Stage: Total product also decreases and marginal product becomes negative.

- **Causes of the Operation of Law of Variable Proportion: (i)** Indivisibility of factors, **(ii)** Division of Labour or specialisation, **(iii)** Imperfect substitutes, **(iv)** Change in factor ratio.
- **Can the Law of Diminishing Returns be Postponed?:** Improvement in technique of production and discovery of fixed factor substitute can postpone the operation of Law of Diminishing Returns for some time, but ultimately it will apply.
- **Causes of Increasing Returns to a Factor: (i)** Full utilisation of fixed factors of production, **(ii)** Better coordination between fixed and variable factors, **(iii)** Improved efficiency of variable factors
- **Causes of Diminishing Returns to a Factor: (i)** Presence of fixed factors, **(ii)** Poor coordination between fixed and variable factors, **(iii)** Imperfect substitution of variable factors
- **Relationship between Average Product (AP) and Marginal Product (MP):**
  - AP increases when MP is greater than AP.
  - AP is at its maximum when both MP and AP are equal.
  - AP decreases when MP is less than AP.
  - MP can be positive, zero or negative, but AP is always positive.
- **Relationship between Total Product (TP) and Marginal Product (MP):**
  - When MP increases, TP increases at an increasing rate.
  - When MP decreases, TP increases at a diminishing rate.
  - When MP is constant, TP increases at a constant rate.
  - When MP is zero, TP is maximum.
  - When MP is negative, TP declines.
- **Point of inflexion**: It is the point upto which the TP increases at an increasing rate and then increases at a diminishing rate. In other words, it is the point at which the slope of TP curve changes.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. State the meaning of law of diminishing returns to a factor.

Ans. The law of diminishing returns to a factor can be defined as the no. of units of variable factor employed are increased then after sometime Total product starts increasing at decreasing rate and then MP starts falling.

Q.2. State the meaning of law variable proportion? [OEB]

Ans. Law of Variable Proportion states that as more and more of the variable factor is combined with the fixed factor, a stage must ultimately come when marginal product of the variable factor starts declining.

Q.3. When total product increases at an increasing rate, what happens to marginal product? [OEB]

Q.4. When total product increases at a decreasing rate, what happens to marginal product? [OEB]

Ans. Marginal Product should be decreasing.

Q.5. Intiyaz saw that the total product had started to decline even after employing more factors of production. Which stage of returns to factor is he in? [OEB]

Ans. Intiyaz is in the stage of negative returns to factor.

Q.6. State giving reason whether the following statement is true or false: [OEB]

(a) "Law of variable proportion states that as we increase more and more units of fixed factor, total product first increases at increasing rate, then at decreasing rate and then starts falling."

Ans. False

Explanation: The given statement is false because as the no. of units of variable factor employed are increased then total product first increases at increasing rate, then at decreasing rate and then starts falling. Also, marginal product first rises, then falls and then becomes negative.

(b) "Law of variable proportion is applicable in long run."

Ans. False

Explanation: The given statement is false because the law of variable proportion or short-run production function is applicable in the short-run and not the long-run.

2 Marks Questions

Q.1. Negative returns cause losses to a firm. What are the causes for Negative returns? [ISC Specimen Paper 2023-24]

Ans. The main reasons for occurrence of negative returns to a factor are:

(i) Limitation of Fixed Factor: The negative returns to a factor apply because some factors of production are of fixed nature, which cannot be increased with increase in variable factor in the short run.

This question is for practice and its solution is available at the end of the chapter

(ii) Poor Co-ordination between Variable and Fixed Factor: When variable factor becomes too excessive in relation to fixed factor, then they obstruct each other. It leads to poor co-ordination between variable and fixed factor. As a result, total output falls instead of rising and marginal product becomes negative.

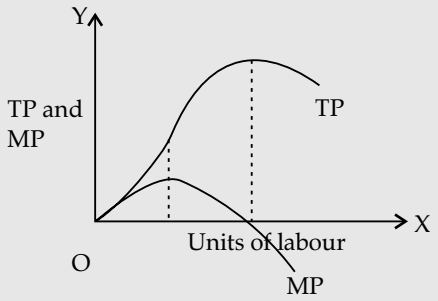
(iii) Decrease in Efficiency of Variable Factor: With continuous increase in variable factor, the advantages of specialisation and division of labour start diminishing. It results in inefficiencies of variable factor, which is another reason for the negative returns to eventually set in. **[Any Two]**

Q.2. With the help of a diagram, state the behaviour of MP when: [ISC 2019]

(a) TP of the variable factor reaches maximum.

(b) TP of the variable factor falls.

Ans.



(a) MP = 0 when TP is maximum.

(b) MP becomes negative when TP is falling.

[ISC Marking Scheme, 2019]

Examiner's Comment

- Majority of the candidates made an improper diagram. Most of the candidates did not mention the variable factor along the x-axis. Some candidates were confused between TP and TU.

Answering Tip

- Learn the meaning and relation between TP and MP.

Q.3. State the assumptions of the law of variable proportions. [OEB]

Q.4. The following headline appeared in the newspaper: "When number of variable factors employed increases from 0,1,2,… then total output increases." Analyse the above given statement and write down the behaviour of total output? [OEB]

Ans. In the short-run production function, when number of variable factor employed increases from 0,1,2,…… keeping fixed factor constant then Total output starts increasing at increasing rate, then at decreasing rate and then starts falling.

Q.5. Rita said to Sita that "The law which operates in the short run is Law of returns to scale whereas the law which operates in the long run is the law of returns to a factor."

Analyse the above statement with reference to the context that whether the statement said by Rita is correct or not? [OEB]

Ans. The above statement said by Rita is wrong because the law which operates in the short run is law of returns to a factor or short-run production function and the law which operates in the long run is called as law of returns to scale or long-run production function.

3 Marks Questions

Q.1. Explain two causes of increasing returns to a factor. [ISC 2017]

Ans. Causes of Increasing Returns to a Factor:

(i) Full utilisation of fixed factors of production.

(ii) Better coordination between fixed and variable factors.

(iii) Improved efficiency of variable factors.

Q.2. Which stage of the variable proportions is considered to be optimum? Why? [OEB]

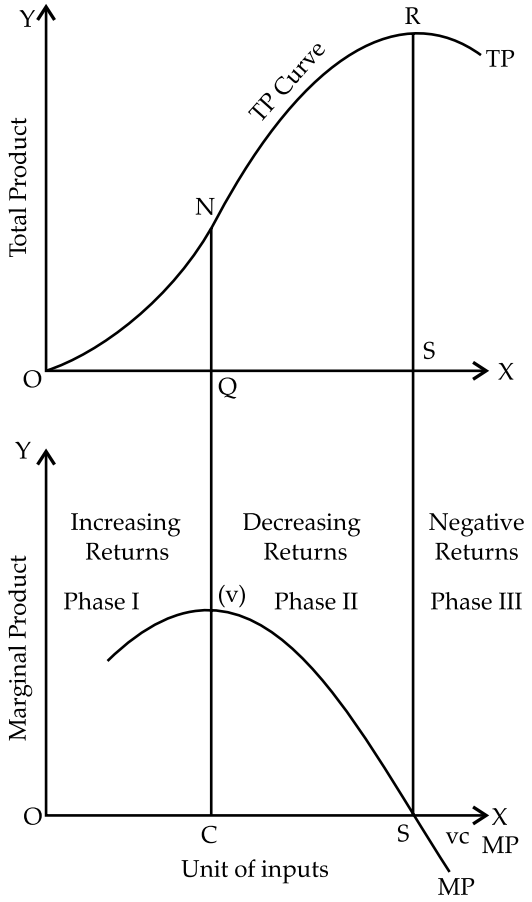
Ans. According to the law of variable proportion, TP first increases at an increasing rate, then at decreasing rate and then starts falling. In the second stage, MP is decreasing but is still positive. This means that TP is still increasing. So, a rational producer would prefer to operate in the second stage of the law of variable proportions.

Q.3. What do you mean by returns to variable factors? [OEB]

Q.4. With the help of a diagram show the effect of Total Product on Marginal Product of a variable factor, when Total Product reaches its maximum. [OEB]

This question is for practice and its solution is available at the end of the chapter

Ans.



Q.5. From the following, identify the three phases of the law of variable proportion. Give reason. [OEB]

Variable units	Marginal Product (in units)
1	20
2	25
3	21
4	16
5	10
6	-2

Ans.

Variable units	Marginal Product (in units)	Phases
1	20	Phase:1
2	25	Increasing returns to a factor
3	21	Phase:2
4	16	Diminishing returns
5	10	to a factor
6	-2	Phase:3
		Negative returns to a factor

4 Marks Questions

Q.1. From the following, identify the three phases of the law of variable proportion. Give reason. [OEB]

Variable units	Total Product (in units)
1	6
2	13
3	17
4	19
5	16

Ans.

Variable units	Total Product (in units)	Marginal Product (in units)	Phases
1	6	6	Phase:1
2	13	7	Increasing returns to a factor
3	17	4	Phase:2
4	19	2	Diminishing returns to a factor
5	16	-3	Phase:3 Negative returns to a factor

Q.2. From the following, identify the three phases of the law of variable proportion. Give reason. [OEB]

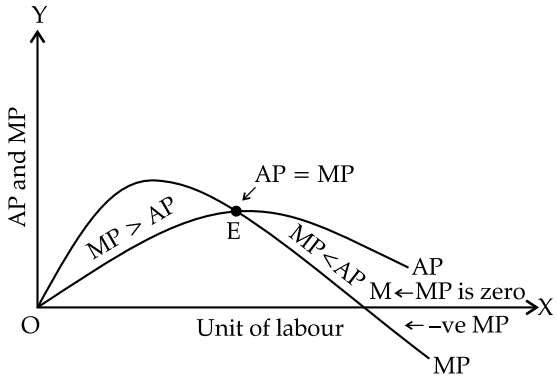
Variable units	Total Product (in units)
1	10
2	22
3	31
4	37
5	32

Ans.

Variable units	Total Product (in units)	Marginal Product (in units)	Phases
1 2	10 22	10 12	Phase:1 Increasing returns to a factor
3 4	31 37	9 6	Phase:2 Diminishing returns to a factor
5	32	–5	Phase:3 Negative returns to a factor

Q.3. How is the Average Product related to the Marginal Product? Explain with the help of a diagram. [OEB]

Ans. The relationship between AP and MP can be explained with the help of the following diagram:



In the diagram:

(i) The AP increases when MP is greater than AP.

(ii) The AP is at its maximum when both MP and AP are equal. This is shown at point E.

(iii) The AP decreases when MP in less than AP.

(iv) MP can be positive, zero or negative, but AP is always positive.

Q.4. Differentiate between Increasing Returns to factor and Decreasing Returns to factor. [OEB]

6 Marks Questions

Q.1. Explain the Law of Variable Proportions with the help of a diagram.

[ISC Specimen Paper 2023-24] [ISC 2015]

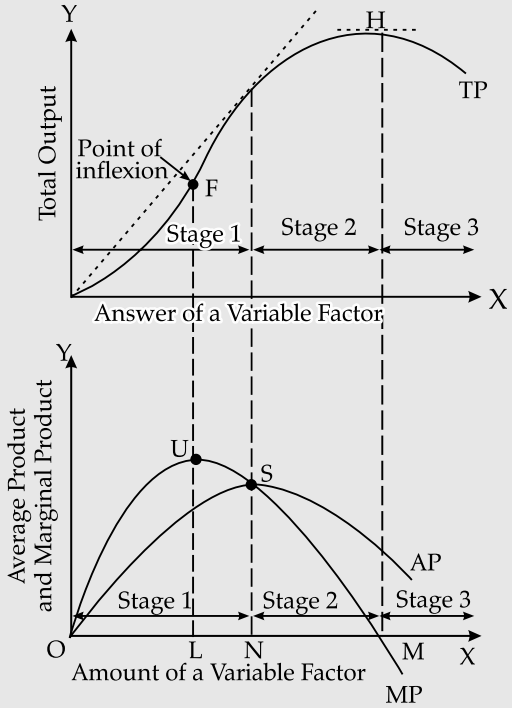
OR

Discuss the three stages of the Law of Variable Proportions with the help of diagrams. In which stage will a rational producer like to operate? [ISC 2020]

OR

If more variable factors are employed to fixed factors, the total product increases initially at increasing rate and finally it decreases. Explain this law with the help of a diagram. [2018]

Ans. The Law of Variable Proportion states that as more and more units of a variable factor are applied to the given quantity of fixed factor total product may increase at an increasing rate initially but eventually it will increase at a diminishing rate.



This question is for practice and its solution is available at the end of the chapter

Stage	TP	MP	AP
I. (increasing return to variable factor)	Initially increase at an increasing rate and then after F at a decreasing rate.	First increases, reaches maximum at point of inflection then starts decreasing.	Increases and reaches maximum. AP < MP
II. (diminishing return to a variable factor)	Continues to **increase at a diminishing rate** and eventually reaches maximum.	**Continues to decrease** and eventually becomes zero (when TP reaches maximum).	From maximum it begins to decrease. **AP > MP**
III. (negative returns)	**Diminishes**	Is negative.	Continues to decrease but is always **positive**.

A rational producer will produce in stage II.

[ISC Marking Scheme, 2020]

Examiner's Comments

The common errors made by the candidates were:

- Not having written the statement of the law.
- Either not indicating the stages in the diagram or wrongly indicating them, particularly the first stage which ended at the point of inflection (where MP is maximum).
- Not showing the proper point of intersection of MP and AP curves.
- Not having written about AP while identifying the zones of production (It is not possible to identify the zones without AP)
- Writing the explanation based on an incorrect diagram.
- Drawing the correct diagram, but not explaining the behaviour of TP, AP and MP for all three stages.
- Discussing the relation between TP - MP and AP -MP.
- Labelling TP, AP and MP curves as TC, AC and MC or TR, AR and MR. and then basing the explanation on the incorrect curves.

Answering Tip

- The students should understand that explanation of any theory without its statement, is incomplete. The law must be stated for any question relating to it. Zones of production must be learnt showing where it starts and where it ends, especially with the AP curve. For adequate practice, students should draw the diagram repeatedly. Relationship between TP, AP and MP must be taught with the help of a table which can be explained in writing. Students should learn the statement and the diagram for Law of Variable Proportions. The correct area on the diagram where the three stages start and end should also be understood. The students need to understand the behaviour of TP, AP and MP under increasing returns, diminishing returns and negative returns stages. They should know that law deals with the behaviour of TP, AP and MP. The students should analyse that a rational producer will operate in stage II.

Q.2. Explain with the help of a diagram the relationship between total product and marginal product.

[ISC 2016]

Ans. The relationship between total product and marginal product:

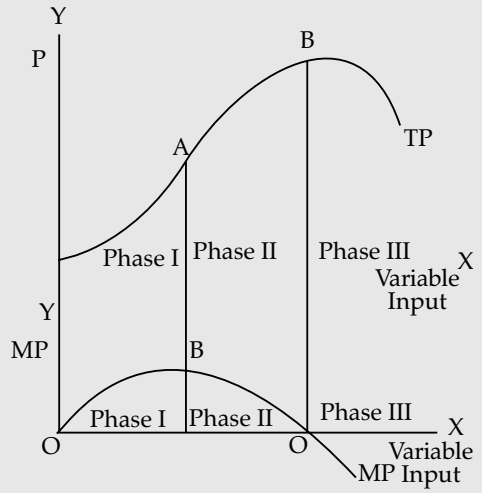
Relationship between TP and MP:

(i) When TP increases at an increasing rate, MP increases. MP is maximum at the point of inflexion.

(ii) TP increases at diminishing rate, MP falls but is positive.

(iii) When TP is maximum, MP is zero.

(iv) When TP falls, MP becomes negative.



[ISC Marking Scheme, 2016]

Examiner's Comments

- **TP curve not drawn correctly by many candidates. Common errors made by candidates were as follows:**

- Candidate did not show the point of inflexion on TP and hence the maximum point of MP was missed out.
- Shape of MP curve was not in conformity with TP Curve.
- Instead of TP and MP relationship, mere explanation of law of variable proportions was written, stage wise.

Answering Tip

- Draw the diagram and explain the relationship of the two curves step wise. Apart from learning the law of variable proportions, the other probable relationship of different curves should also be analysed and understood.

OBJECTIVE TYPE QUESTIONS

[A] Multiple Choice Questions

Q.1. When the Marginal Product turns negative, Total Product will. **[ISC 2023]**

(A) fall **(B)** rise
(C) remain constant **(D)** be zero

Ans. Option (A) is correct.

Explanation: According to the law of variable proportion or short-run production function, when marginal product turns negative, total product will fall, while average product will remain positive.

Q.2. Which of the following equation is correct? **[ISC Semester-I, 2022]**

(A) $MP = TP_{n-TPn-1}$ **(B)** $MP = \frac{AP}{L}$
(C) $MP = \frac{TP}{L}$ **(D)** $MP = TP - AP$

Q.3. If some inputs are variable and some inputs are fixed, then the functional relation is a case of **[ISC Semester-I, 2022]**

(A) Short run production function.
(B) Long run production function.
(C) Law of variable proportions.
(D) Diminishing marginal utility.

Ans. Option (A) is correct.

Explanation: Short-run production function refers to a situation when only one input is variable and all other inputs are assumed to be constant. The behaviour of output when only one input is changed and other inputs are fixed are termed as returns to a factor. It is studied as law of variable proportions or laws of returns.

Q.4. A rational producer will operate in _____ stage of law of variable proportions. **[ISC Semester-I, 2022]**

(A) Stage I **(B)** Stage II
(C) Stage III **(D)** Stage IV

Ans. Option (B) is correct.

Explanation: According to the law of variable proportion, TP first increases at an increasing rate, then at decreasing rate and then starts falling. In the second stage, MP is decreasing but is still positive. This means that TP is still increasing. So, a rational producer would prefer to operate in the second stage of the law of variable proportions.

Q.5. If MP of labour is below AP of labour the **[ISC Semester-I, 2022]**

(A) MP is negative **(B)** MP = 0
(C) AP is falling **(D)** AP is negative

Ans. Option (C) is correct.

Q.6. The law of variable proportion states:

When a variable factor is added to fixed factor, the total product increases at increasing rate and finally total product _____. **[ISC Semester-I, 2022]**

(A) Rises
(B) Falls
(C) Remains the same
(D) Rises at diminishing rate

Ans. Option (D) is correct.

Explanation: The law of variable proportions states that as more and more of the variable factor is combined with the fixed factor, marginal product (MP) of the variable factor may initially increase at an increasing rate and then increase at a diminishing rate, and subsequently stabilise but must finally decrease.

Q.7. In the case of the law of variable proportions, diminishing returns to a factor occur when _____. **[ISC Semester-I, 2022]**

(A) Units of variable inputs are added to a fixed factor and AP and MP fall.
(B) Units of variable inputs are added to fixed factor, AP increases.
(C) Size of plant increases in the long run, and MP is constant.
(D) Quality of the fixed factor increases and returns to variable input falls.

Ans. Option (A) is correct.

Explanation: In the second phase, every additional variable factor adds lesser and lesser amount of output. It means TP increases at a diminishing rate and MP falls with increase in variable factor. That is why this phase is known as diminishing returns to a factor.

This question is for practice and its solution is available at the end of the chapter

Q.8. When AP reaches maximum, TP is. **[ISC Semester-I, 2022]**

(A) Also maximum.
(B) Increases at diminishing rate
(C) Zero
(D) Negative

Ans. Option (B) is correct.

Explanation: When AP reaches maximum, total product is rising but at a decreasing rate.

Q.9 One of the reasons for negative returns to factor is: **[ISC Semester-I, 2022]**

(A) Imperfect substitutability.
(B) Overcrowding and management problems.
(C) Disturbance in optimum proportion.
(D) Indivisibility of factors.

Ans. Option (B) is correct.

Explanation: Negative returns arises due to the fact that the amount of variable factors has become excessive of the fixed factor. This leads to overcrowding and chaos and hampers each other's work.

Q.10. The shape of MP curve is _____ shape. **[ISC Semester-I, 2022]**

(A) U **(B)** Inverse U
(C) Inverse S **(D)** S

Q.11. When 50 hours of labour are spent, total output produced is 2000 units and when 55 hours of labour are spent, total output produced is 2,250 units. In this case MP is **[ISC Semester-I, 2022]**

(A) 500 **(B)** 40
(C) 45 **(D)** 50

Ans. Option (D) is correct.

Explanation: MP for 5 hours = 2,250 – 2,000 = 250

$$\text{MP of 1 hour} = \frac{250}{5} = 50$$

Q.12. In the stage of diminishing returns of a variable factor. **[ISC Semester-I, 2022]**

(A) TP is increasing, AP is increasing, MP is also increasing.
(B) TP is constant, AP is decreasing, MP is also increasing.
(C) TP is increasing, AP declines, MP is declining to become zero.
(D) TP is decreasing, AP is decreasing, MP is negative.

Ans. Option (C) is correct.

Q.13. Long run production function studies the change in output when. **[ISC Semester-I, 2022]**

(A) One input is variable.
(B) One input is fixed.
(C) All inputs are variable.
(D) All inputs are fixed.

Ans. Option (C) is correct.

Explanation: In long run production function all the factors of production are variable and are subject to change with the change in output.

[B] Assertion Reason Questions

Directions: In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.1. Assertion (A): Average product increases only when marginal product increases.

Reason (R): AP increases so long as MP is more than AP, whether MP is rising or falling. **[OEB]**

Ans. Option (D) is correct.

Explanation: AP increases when MP is greater than AP. AP is at its maximum when both MP and AP are equal. AP decreases when MP in less than AP. MP can be positive, zero or negative, but AP is always positive.

Q.2. Assertion (A): Increasing returns to a factor is a short run phenomenon.

Reason (R): Greater application of the variable factor ensures fully utilisation of the fixed factor. **[OEB]**

Ans. Option (B) is correct.

Explanation: Increasing returns to a factor is a short run phenomenon as in the long run it is a constant proportion type production function in which factor ratio remains constant, no matter what the level of output is.

Q.3. Assertion (A): MP increases in stage I as per law of variable proportion.

Reason (R): Fuller use of the variable factor leads to the Marginal Product to increase. **[OEB]**

Ans. Option (C) is correct.

Explanation: In the first phase, every additional variable factor adds more and more to the total output. It means TP increases at an increasing rate and MP of each variable factor rises.

Q.4. Assertion (A): Average product can never be negative.

Reason (R): Average product is the Total Product divided by total variable input. **[OEB]**

Ans. Option (A) is correct.

Explanation: Average product can never be negative as when we divide the total product with the variable input, it can never be negative.

Q.5. Assertion (A): Law of Diminishing Returns can be postponed.

Reason (R): Improvement in technique of production and discovery of fixed factor substitute. **[OEB]**

Ans. Option (A) is correct.

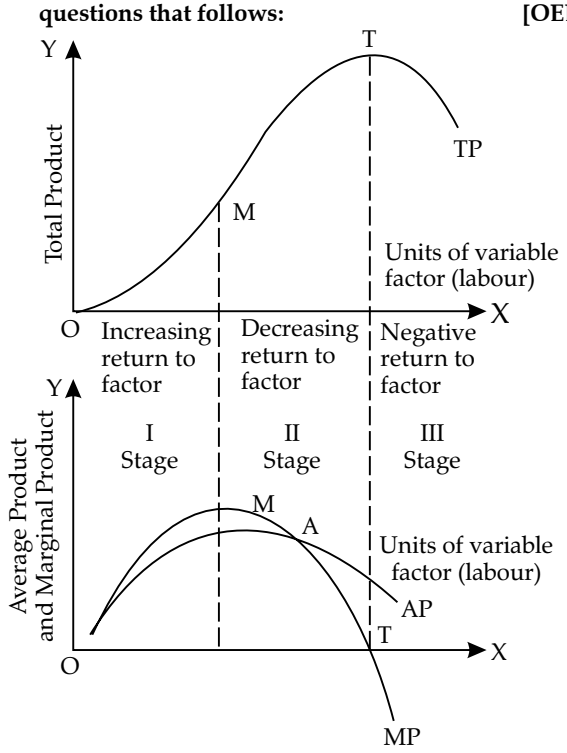
Explanation: Improvement in technique of production and discovery of fixed factor substitute can postpone the operation of Law of Diminishing Returns for some time, but ultimately it will apply.

This question is for practice and its solution is available at the end of the chapter

Case Based Questions

[A] Multiple Choice Questions

I. Study the graph given below and answer the questions that follows: [OEB]



(a) What is the above graph depicting?

(A) Law of Variable Proportions.
(B) Consumer's Equilibrium.
(C) Producer's Equilibrium.
(D) Law of Returns to Scale.

Ans. Option (A) is correct.

(b) Which stage is considered to be the most optimal stage?

(A) Increasing Return to Factor.
(B) Decreasing Return to Factor.
(C) Negative Return to Factor.
(D) Both (A) and (B).

Ans. Option (B) is correct.

Explanation: According to the law of variable proportion, TP first increases at an increasing rate, then at decreasing rate and then starts falling. In the second stage, MP is decreasing but is still positive. This means that TP is still increasing. So, a rational producer would prefer to operate in the second stage of the law of variable proportions.

(c) Which of the following will lead the producer to be in the Stage III?

(A) More labours as compared to Capital.
(B) Reduction in the managerial efficiency.
(C) Overutilisation of the production capacity.
(D) All of the above.

Ans. Option (D) is correct.

(d) What can be said about the MP in the Stage III?

(A) Reaches the maximum
(B) Reaches zero
(C) Increases
(D) Becomes negative

Ans. Option (D) is correct.

Explanation: At stage III, the TP starts declining, AP declines but remains positive and MP falls below zero and becomes negative.

[B] Subjective Based Questions

II. Law of Variable Proportions occupies an important place in economic theory. This law is also known as Law of Proportionality.

Keeping other factors fixed, the law explains the production function with one factor variable. In the short run when output of a commodity is sought to be increased, the law of variable proportions comes into operation.

Therefore, when the number of one factor is increased or decreased, while other factors are constant, the proportion between the factors is altered. For instance, there are two factors of production viz., land and labour.

Land is a fixed factor whereas labour is a variable factor. Now, suppose we have a land measuring 5 hectares. We grow wheat on it with the help of variable factor i.e., labour. Accordingly, the proportion between land and labour will be 1: 5. If the number of labourers is increased to 2, the new proportion between labour and land will be 2: 5. Due to change in the proportion of factors there will also emerge a change in total output at different rates. This tendency in the theory of production called the Law of Variable Proportion. **[OEB]**

- Source: economicsdiscussion.net

(a) What is the above extract talking about? Explain.

Ans. The above extract is talking about the Law of Variable Proportion.

Law of variable proportions states that when total on output or production of a commodity is increased by adding units of a variable input, while the quantities of other inputs are held constant, then MP of the variable factor initially rises and then falls after reaching a certain level of employment of the variable factor.

(b) What are the various stages operating in the law as stated above?

Ans. The stages are:

(i) Increasing Returns to Factor.
(ii) Decreasing Returns to Factor.
(iii) Negative Returns to Factor.

(c) How is Marginal Product affected from the law as stated above?

Ans. This law states that the total physical product (TP) first increases at an increasing rate and finally falls, when one variable factor output is changed keeping all other factors constant. Behaviour : It is seen that the marginal product (MP) first rises, then falls and finally becomes negative.

(d) Why does the firm operate in the first stage of the law?

Ans. There are three important reasons for the operation of increasing returns to a factor:

(a) Better Utilisation of the Fixed Factor: In the first phase, the supply of the fixed factor (say, land) is too large, whereas variable factors are too few. So, the fixed factor is not fully utilised. When variable factors are increased and combined with fixed factor, then fixed factor is better utilised and output increases at an increasing rate.

(ii) Increased Efficiency of Variable Factor: When variable factors are increased and combined with the fixed factor, then former is utilised in a more efficient manner. At the same time, there is greater cooperation and high degree of specialisation between different units of the variable factor.

(iii) Indivisibility of Fixed Factor: Generally, the fixed factors which are combined with variable factors are indivisible. Such factors cannot be divided into smaller units. Once an investment is made in an indivisible fixed factor, then addition of more and more units of variable factor improves the utilisation of fixed factor. The increasing returns apply as long as optimum level of combination between variable and fixed factor is achieved.

(e) Which stage is considered the most optimal stage? Why?

Ans. According to the law of variable proportion, TP first increases at an increasing rate, then at decreasing rate and then starts falling. In the second stage, MP is decreasing but is still positive. This means that TP is still increasing. So, a rational producer would prefer to operate in the second stage of the law of variable proportions.

III. Production theory in economics refers to how businesses decide the quantities of outputs to produce in response to demand. The firm produces goods or commodities with limited resources, and this means that even if it somehow knows people are willing to buy whatever quantity of goods it produces, it just can't instantly make an unlimited quantity of goods.

Production theory in economics refers to how businesses decide the quantities of outputs to produce in response to demand.

The resources firms use in production are called the factors of production, and they are also known as inputs.

Let's say the firm uses labour and land as its two resources to make the goods. At any point in time, the firm can increase or decrease the number of workers it employs in response to an increase or decrease in demand. Since the number of workers is so easy to change at any point in time, economists refer to it as a variable factor of production or input. On the other hand, since land cannot be changed so easily, it is often called a fixed input.

Since the production theory is about deciding how much output to make, the production function helps by showing how the output will change when a variable input changes. For example, how many products will the company make if it employs two more employees? The production function answers this question! **[OEB]**

- The Theory of Production: Examples & Function | Study Smarter

(a) What is production theory?

Ans. Production theory is the decision made by the business firms in order to use the various factors of production in order to create utility and respond to demand.

(b) Name the various factors or production.

Ans. The four factors of production are land, labour, capital and entrepreneur.

(c) Define Production function. State which factors are considered to be fixed and variable in case of the short-run.

Ans. Production function is the functional relationship between physical input and physical output. In production function, only physical inputs are taken into consideration in relation to the physical output and the market prices are not included in it.

$$Qx = f\{L,K\}$$

Labour is considered to be variable and Land and Capital are considered to be fixed in the short run.

(d) Name and define the types of production function.

Ans. The two types of production function are:

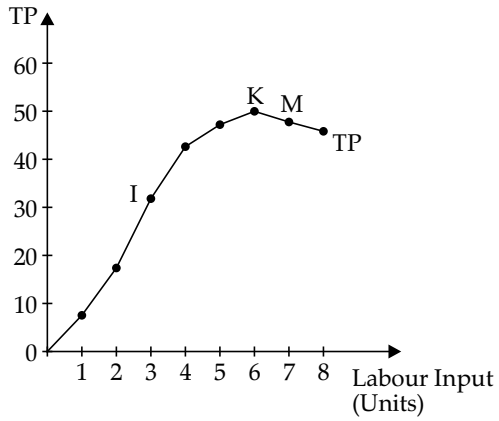
(i) Short Run Production Function or Returns to a Factor: It is related to short-run, when only one factor of production is variable and others are fixed. Since units of variable factors are increased continuously, so factor ratio changes in this production function. So, law of return to a factor is applied. It is also called variable proportion type production function.

(ii) Long Run Production Function or Returns to Scale: It is related to long-run and all the factors are changed in same ratio. In this case, proportional relation between production and factors of production in the long run is called Returns to Scale. Here, scale of production changes. So, law of returns to scale is applied. It is also called return to scale.

(e) Explain the production function with the help of a hypothetical example.

Ans. A Short Run Production Function is expressed as Qx = f(LK). Here L = Labour and K = Capital. The behaviour of Total Product in this function is explained below:

Units of Labour	Total Product	Marginal Product	Change in the TP Curve
0	0	-	TP rises at an increasing rate till point I
1	7	7	
2	18	11	
3	33	15	
4	44	11	TP rises at a decreasing rate from point I to K
5	48	4	
6	51	3	TP attains maximum and becomes constant from point K to M
7	51	0	
8	49	-2	TP falls after M



From the figure and the schedule, we can analyse that initially as more and more labour units are employed, the TP curve increases at an increasing rate till point I (corresponding to 2 units of labour units). After point I, the TP curve rises at a decreasing rate till the point K. The point I is also known as point of inflection. This is because passing through the point I, the curvature of the TP curve changes from convex to concave. With successive rise in the labour units, the TP curve continues to rise and attains its maximum point K (corresponding to 6 units of labour) and remains stable till point M. Beyond this point, the TP curve starts falling when more than 8 units of labour are employed.

Solutions for Practice Questions (Topic-1)

1 Mark Questions

2. The process in which raw material is converted into final goods is called as production. Production adds value to the commodity.

2 Marks Questions

2. Production function: It is a functional relationship between output and inputs.
$Q = f$ (inputs) $Q_x = f(f_1, f_2, f_3.....)$ or A production function shows the maximum quantity of a commodity that can be produced per unit of time with the given amount of inputs when the best production technique available is used.

[ISC Marking Scheme, 2020]

Examiner's Comment

- Many candidates did not write the terms 'functional' and 'best production' technique while defining production function. Some candidates were unable to establish the functional relationship between input & output.

Answering Tip

- The students need to understand the difference between production (as process) and production function (Input- Output relationship). The significance of the term 'functional/ mathematical/ technological' should be clear to the students. The students should also ensure that they write $Q = f(f_1, f_2, ------f_n)$.

3 Marks Questions

4. Differences between Short Run and Long Run Production Function are:

S. No.	Long Run Production Function	Short Run Production Function
(i)	It is a constant proportion type production function in which factor ratio remains constant, no matter what the level of output is.	It is a variable proportion type production functionin which factor ratio changes with change in the level of output.
(ii)	Output is sought to be increased by increasing the application of all factors.	Output is sought to be increased by increasing the application of one variable factor only.
(iii)	Scale of production tends to change with the change in the level of output.	Scale of production remains constant, even when the level of output changes.
(iv)	Long period production function is studied with reference to 'Returns to Scale'.	Short period production function is studied with reference to 'Returns to a Factor'.

Solutions for Practice Questions (Topic-2)

1 Mark Questions

3. Marginal Product should be increasing.

2 Marks Questions

3. **The Law of Variable Proportion is based on these assumptions:**
   (i) The firm operates in the short run.
   (ii) Factors can be substituted mutually
   (iii) Technology is constant
   (iv) All units of variable factors are homogenous.

   (Any Two)

3 Marks Questions

3. Law of returns to variable factor states that when total output or production of a commodity is increased by adding units of a variable input, while the quantities of other inputs are held constant, then MP of the variable factor initially rises and then falls after reaching a certain level of employment of the variable factor.

4 Marks Questions

4. **Increasing returns to factor:** In this stage, TP increases at an increasing rate and MP rises till it reaches its maximum point.

   Decreasing returns to factor: In this stage, TP increases at a decreasing rate and reaches maximum, and MP becomes zero.

   Causes of Increasing Returns to a Factor: (i) Full utilization of fixed factors of production, (ii) Better coordination between fixed and variable factors, (iii) Improved efficiency of variable factors

   Causes of Diminishing Returns to a Factor: (i) Presence of fixed factors, (ii) Poor coordination between fixed and variable factors, (iii) Imperfect substitution of variable factors.

Solutions for Objective Type Questions

[A] Multiple Choice Questions

2. **Option (A) is correct.**
   Explanation: Marginal product is the change in the total output with every additional input used. Thus $MP = TP_n - TP_{n-1}$.

10. **Option (B) is correct.**

REFLECTION

(a) Do you understand the difference between short run production function and long run production function?

(b) Do you know the meaning of Total Product, Marginal Product and Average Product and can you calculate them?

(c) Can you state the law of variable proportions and the various stages in it?

(d) Can you explain the reasons for each stage of variable proportions and state with reason which stage is optimal?